

Accepted Manuscript

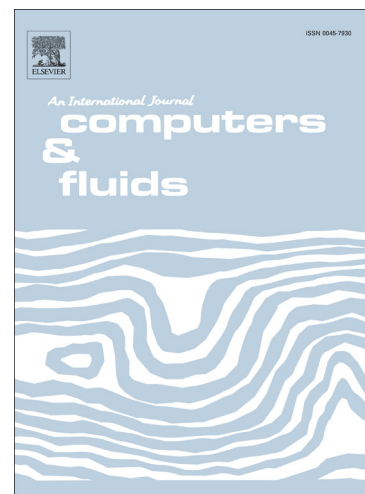
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PII: S0045-7930(14)00362-4
DOI: <http://dx.doi.org/10.1016/j.compfluid.2014.09.025>
Reference: CAF 2677

To appear in: *Computers & Fluids*

Received Date: 26 March 2014
Revised Date: 4 August 2014
Accepted Date: 3 September 2014



Please cite this article as: Nonomura, T., Terakado, D., Abe, Y., Fujii, K., A New Technique for Freestream Preservation of Finite-difference WENO on Curvilinear Grid, *Computers & Fluids* (2014), doi: <http://dx.doi.org/10.1016/j.compfluid.2014.09.025>

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A New Technique for Freestream Preservation of Finite-difference WENO on Curvilinear Grid

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Abstract

A new technique for a finite-difference weighted essentially nonoscillatory scheme (WENO) on curvilinear grids to preserve freestream is introduced. This technique first divides the standard finite-difference WENO into two parts: 1) a consistent central difference part and 2) a numerical dissipation part. For the consistent central difference part, the conservative metric technique is directly adopted. For the numerical dissipation part, it is proposed that the metric term should be frozen for constructing the upwinding flux. This treatment only affects the numerical dissipation part, and the order of accuracy is maintained. With this technique, the freestream is perfectly preserved, and the flow fields are better resolved on wavy and random grids.

1 Introduction

Direct numerical simulations and large eddy simulations have recently become essential tools for understanding unsteady flow phenomena. A problem with these simulations is that they are computationally very expensive because they require a very fine computational grid to maintain the spatial accuracy of their results. One possible solution is to use high-order schemes with fewer grid points. A high-order scheme has higher resolution than a low-order scheme, and it can save on computational time by a factor of 100 in three-dimensional problems. Moreover, according to Shu[29], when a high-order scheme is implemented in a multidimensional problem, finite-difference schemes cost one-tenth as much as finite-element or volume methods. This is because the finite-difference scheme can be constructed using a dimension-by-dimension procedure while finite-element or volume methods require multidimensional reconstruction procedures.

Therefore, various high-order finite-difference schemes have been proposed. Examples include the compact difference scheme proposed by Lele[21] and applied to the generalized coordinate system by Visbal and Gaitonde[34], the finite-difference version of the weighted essentially nonoscillatory scheme (WENO) proposed by Jiang and Shu[17], and the weighted compact nonlinear scheme (WCNS) proposed by Deng and Zhang[11]. Here WENO and WCNS have been developed to solve flow fields that include discontinuities, such as shock waves, without numerical oscillations.

When finite-difference schemes are applied to curvilinear coordinate systems, which are often used for practical simulations, the surface closure law (SCL) identity[37] which is one of the geometric conservation law (GCL) identity is important. If the SCL identity is not satisfied, errors are generated from the freestream on curvilinear grids. This type of error hides very small oscillations, such as turbulent flow structures or aeroacoustic waves, and sometimes makes the simulation unstable. With regard to SCL, there are a couple of techniques for low-order schemes, such as the averaging and differencing technique by Pulliam and Stegar,[27] the conservative metric technique by Thomas and Lombard,[31] and the finite-volume evaluation by Vinokur.[32] Among these techniques, the conservative metric technique can be adopted for high-order central difference schemes, as demonstrated by Visbal and Gaitonde.[34] This conservative metric was extended to the symmetric conservative metric by Vinokur and Yee.[33] It is also extended to the time metrics for the volume conservation law (VCL) which is another law in GCL.[1] Using linear high-order finite-difference schemes, the discretization of the symmetric conservative forms of the spatial and temporal metrics are interpreted to be a linear combination of the cell surface areas and volumes around the computational node, respectively, by Deng et al.[10] and Abe et al.[2]

With regard to SCL in weighted high-order schemes, Cai and Ladeinde[7] and Nonomura et al.[24,25] demonstrated that the standard finite-difference (flux version) WENO scheme cannot satisfy the SCL identity with direct implementation into a body-fitted coordinate system even by using the conservative metric technique. On the other hand, WCNS, a variant of WENO, can preserve the freestream by independent interpolations of flow variables and metrics, that can be adopted owing to the unique definition of the numeri-

cal flux, as discussed in the Reference.[25] The fact that WCNS can preserve the freestream was later confirmed by Deng et al.[9] The similar idea was also proposed as the point flux WENO (PFWENO) scheme.[20] More recently, Jiang et al.[19,18] proposed an alternative finite-difference form of WENO (AWENO) for freestream preservation, where alternative form for essentially nonoscillatory scheme was first proposed by Shu and Osher.[30] The idea of cancellation of error in these forms is essentially the same as that in WCNS: it adopts the independent interpolation of flow variables and metrics. Actually, the present authors[3] shows that the AWENO scheme is analytically equivalent to a variant of WCNS: a special case of a hybrid weighted compact nonlinear scheme.[9] One solution for satisfying the SCL in weighted schemes is the use of WCNS, PFWENO or AWENO scheme, which are not standard form of the finite-difference WENO scheme. However, the standard WENO scheme has often been modified and there are many special versions. Therefore, an extension of the standard WENO scheme to satisfy SCL also appears to be worth considered.

In this study, a technique is proposed for a finite-difference WENO scheme to preserve the freestream on stationary curvilinear grids. This technique first divides the finite-difference WENO scheme into the consistent central differencing part and the numerical dissipation part. Then, the appropriate treatment is adopted for each part.

The detailed comparison between the present WENO scheme and WCNS, PFWENO or AWENO scheme (previous technique) is not conducted because the resolution of those schemes even on the Cartesian grid differs and careful discussion seems to be required. The difference in resolution of the standard WENO scheme and WCNS, PFWENO, or AWENO schemes will be reported

in the Reference.[4] However, the limited comparison of the present WENO scheme and WCNS (the previous technique) in terms of resolution and efficiency on a wavy grid is conducted as fairly as possible in the current state. This is described in Appendix B. It should be noted that the freestream preservation for the standard WENO scheme is demonstrated in this paper for the first time.

This paper is organized as follows: Section 2 illustrates the procedure of the original WENO scheme and the WENO scheme with the technique for satisfying the SCL identities. Section 3 describes the test cases of freestream, vortex, and shock flows and flow around a delta wing. Moreover, this section describes the capability and validity of this technique. Finally, Section 4 concludes the present paper.

2 Formulation of WENO on a Curvilinear Coordinate System and a Technique for Freestream Preservation

In this section, the WENO formulation for the Euler equation on a curvilinear coordinate system is introduced.

2.1 Euler Equation on Curvilinear Coordinate System

The Euler equation in Cartesian coordinates is given as follows:

$$\frac{\partial Q}{\partial t} + \frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} + \frac{\partial G}{\partial z} = 0, \quad (1)$$

where

$$\begin{aligned}
 Q &= (\rho \quad \rho u \quad \rho v \quad \rho w \quad \rho e)^T, \\
 E &= (\rho u \quad \rho u^2 + p \quad \rho uv \quad \rho uw \quad u(\rho e + p))^T, \\
 F &= (\rho v \quad \rho uv \quad \rho v^2 + p \quad \rho vw \quad v(\rho e + p))^T, \\
 G &= (\rho w \quad \rho vw \quad \rho vw \quad \rho w^2 + p \quad w(\rho e + p))^T.
 \end{aligned} \tag{2}$$

Here ρ is the density; u , v , and w are the x -, y -, and z -direction velocity components, respectively; e is the total energy per unit mass; and p is the pressure. Here the state of equation for ideal gas is used in this study:

$$\rho e = \frac{p}{\gamma - 1} + \frac{1}{2}\rho(u^2 + v^2 + w^2). \tag{3}$$

The three-dimensional Euler equations in the generalized coordinate system are expressed as

$$\frac{\partial \tilde{Q}}{\partial t} + \frac{\partial \tilde{E}}{\partial \xi} + \frac{\partial \tilde{F}}{\partial \eta} + \frac{\partial \tilde{G}}{\partial \zeta} = 0, \tag{4}$$

where

$$\begin{aligned}
 \tilde{Q} &= \frac{Q}{J}, \\
 \tilde{E} &= \frac{1}{J}(\xi_x E + \xi_y F + \xi_z G), \\
 \tilde{F} &= \frac{1}{J}(\eta_x E + \eta_y F + \eta_z G), \\
 \tilde{G} &= \frac{1}{J}(\zeta_x E + \zeta_y F + \zeta_z G).
 \end{aligned} \tag{5}$$

The standard forms of the Jacobian J and the metrics $\xi_x, \xi_y, \dots$ are expressed as follows:

$$\begin{aligned}
 J^{-1} &= x_{\xi}(y_{\eta}z_{\zeta} - y_{\zeta}z_{\eta}) + x_{\eta}(y_{\zeta}z_{\xi} - y_{\xi}z_{\zeta}) + x_{\zeta}(y_{\xi}z_{\eta} - y_{\eta}z_{\xi}), \\
 \frac{\xi_x}{J} &= (y_{\eta}z_{\zeta} - y_{\zeta}z_{\eta}), \\
 \frac{\xi_y}{J} &= (z_{\eta}x_{\zeta} - z_{\zeta}x_{\eta}), \\
 \frac{\xi_z}{J} &= (x_{\eta}y_{\zeta} - x_{\zeta}y_{\eta}), \\
 \frac{\eta_x}{J} &= (y_{\zeta}z_{\xi} - y_{\xi}z_{\zeta}), \\
 \frac{\eta_y}{J} &= (z_{\zeta}x_{\xi} - z_{\xi}x_{\zeta}), \\
 \frac{\eta_z}{J} &= (x_{\zeta}y_{\xi} - x_{\xi}y_{\zeta}), \\
 \frac{\zeta_x}{J} &= (y_{\xi}z_{\eta} - y_{\eta}z_{\xi}), \\
 \frac{\zeta_y}{J} &= (z_{\xi}x_{\eta} - z_{\eta}x_{\xi}), \\
 \frac{\zeta_z}{J} &= (x_{\xi}y_{\eta} - x_{\eta}y_{\xi}).
 \end{aligned} \tag{6}$$

In this study, the following symmetric conservative form[33] of the metrics is used instead of the above formulation.

$$\begin{aligned}
 J^{-1} &= \frac{1}{3} \left\{ \left(\frac{\xi_x}{J}x + \frac{\xi_y}{J}y + \frac{\xi_z}{J}z \right)_{\xi} + \left(\frac{\eta_x}{J}x + \frac{\eta_y}{J}y + \frac{\eta_z}{J}z \right)_{\eta} + \left(\frac{\zeta_x}{J}x + \frac{\zeta_y}{J}y + \frac{\zeta_z}{J}z \right)_{\zeta} \right\} \\
 \frac{\xi_x}{J} &= \frac{1}{2} \left\{ (y_{\eta}z - z_{\eta}y)_{\zeta} - (y_{\zeta}z - z_{\zeta}y)_{\eta} \right\}, \\
 \frac{\xi_y}{J} &= \frac{1}{2} \left\{ (z_{\eta}x - x_{\eta}z)_{\zeta} - (z_{\zeta}x - x_{\zeta}z)_{\eta} \right\}, \\
 \frac{\xi_z}{J} &= \frac{1}{2} \left\{ (x_{\eta}y - y_{\eta}x)_{\zeta} - (x_{\zeta}y - y_{\zeta}x)_{\eta} \right\}, \\
 \frac{\eta_x}{J} &= \frac{1}{2} \left\{ (y_{\zeta}z - z_{\zeta}y)_{\xi} - (y_{\xi}z - y_{\xi}z)_{\zeta} \right\}, \\
 \frac{\eta_y}{J} &= \frac{1}{2} \left\{ (z_{\zeta}x - x_{\zeta}z)_{\xi} - (z_{\xi}x - x_{\xi}z)_{\zeta} \right\}, \\
 \frac{\eta_z}{J} &= \frac{1}{2} \left\{ (x_{\zeta}y - y_{\zeta}x)_{\xi} - (x_{\xi}y - y_{\xi}x)_{\zeta} \right\}, \\
 \frac{\zeta_x}{J} &= \frac{1}{2} \left\{ (y_{\xi}z - y_{\xi}z)_{\eta} - (y_{\eta}z - y_{\eta}z)_{\xi} \right\}, \\
 \frac{\zeta_y}{J} &= \frac{1}{2} \left\{ (z_{\xi}x - y_{\xi}z)_{\eta} - (z_{\eta}x - x_{\eta}z)_{\xi} \right\}, \\
 \frac{\zeta_z}{J} &= \frac{1}{2} \left\{ (x_{\xi}y - y_{\xi}z)_{\eta} - (x_{\eta}y - y_{\eta}x)_{\xi} \right\}.
 \end{aligned} \tag{8}$$

The above two forms of the metrics are analytically equivalent, but not numerically. Visbal and Gaitonde[34] reported that the linear difference scheme, which is applied to the generalized coordinate system, can preserve the freestream condition if the metric terms in the conservative form and convection terms are evaluated by the unique schemes in each direction. In this study, this technique is referred to as the conservative metric technique. Readers are referred to the previous articles for the detailed conditions and mathematical proof on this conservative metric technique[1,2,9,10,35]

2.2 WENO scheme for Euler Equation on the Curvilinear Coordinate System

The direct extension of fifth-order ($r = 3$) WENO with Lax splitting in a curvilinear coordinate system is explained. First, fluxes are transformed into characteristic forms as follows:

$$\mathcal{E}_{k,m}^{\pm} = l_{j;m} \left(\frac{\xi_{x_k} E_k + \xi_{y_k} F_k + \xi_{z_k} G_k \pm \lambda_m Q}{J_k} \right) \quad (k = j-2, j-1, j, j+1, j+2), \quad (9)$$

where $\mathcal{E}_{k,m}$ denotes the m th characteristic flux, and $l_{j;m}$ denotes the m th left eigenvector of the Roe matrix[28] based on the (j) and $(j+1)$ th grid points. Here $\tilde{E}_k^+$ denotes the upwinding flux using Lax splitting, and $\lambda_{k:m}$ is defined as follows.

$$\lambda = \kappa \max (|\lambda_{j-2,m}|, |\lambda_{j-1,m}| \dots, |\lambda_{j+2,m}|, |\lambda_{ROE,m}|), \quad (10)$$

where $\lambda_{k,m}$ is the eigenvalue of the m th characteristic waves on the k th grid point, and $\lambda_{ROE,m}$ is that computed by the Roe average[28] of the quantities on the (j) and $(j+1)$ th grid points. In this study, this low-dissipative

by Balsara and Shu.[6] For a severe shock problem, $\lambda_{k,m}$ should be replaced by $\sigma_k = \max(\lambda_1, \lambda_2, \dots, \lambda_5)$ for preventing the carbuncle phenomenon. This highly dissipative Lax splitting is only used for the double Mach reflection problem, as discussed later.

Then, the characteristic form of the WENO numerical flux $\tilde{\mathcal{E}}_{j+1/2,m}^{WENO+}$ is constructed as follows:

$$\tilde{\mathcal{E}}_{j+1/2,m}^{WENO+} = w_m^1 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+1} + w_m^2 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+2} + w_m^3 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+3}, \quad (11)$$

where

$$\begin{aligned} \mathcal{E}_{j+1/2,m}^{WENO+1} &= \frac{1}{3} \mathcal{E}_{j-2:m}^+ - \frac{7}{6} \mathcal{E}_{j-1:m}^+ + \frac{11}{6} \mathcal{E}_{j:m}^+, \\ \mathcal{E}_{j+1/2,m}^{WENO+2} &= -\frac{1}{6} \mathcal{E}_{j-1:m}^+ + \frac{5}{6} \mathcal{E}_{j:m}^+ + \frac{1}{3} \mathcal{E}_{j+1:m}^+, \\ \mathcal{E}_{j+1/2,m}^{WENO+3} &= \frac{1}{3} \mathcal{E}_{j:m}^+ + \frac{5}{6} \mathcal{E}_{j+1:m}^+ - \frac{1}{6} \mathcal{E}_{j+2:m}^+, \end{aligned} \quad (12)$$

and w^1 , w^2 , and w^3 are nonlinear weights. These nonlinear weights are determined as follows:

$$w_m^{+k} = \frac{\alpha_{j:m}^{+k}}{\alpha_{j:m}^{+1} + \alpha_{j:m}^{+2} + \alpha_{j:m}^{+3}} (k = 1, 2, 3), \quad (13)$$

where

$$\alpha_{j:m}^{+k} = \frac{C^{+k}}{(IS_{j:m}^k + \epsilon)^p} \quad (k = 1, 2, 3), \quad (14)$$

and

$$\begin{aligned}
 C^{+1} &= \frac{1}{10}, C^{+2} = \frac{6}{10}, C^{+3} = \frac{3}{10}, \\
 IS_m^{+1} &= \frac{1}{4} \left(\tilde{\mathcal{E}}_{j-2:m} - 4\tilde{\mathcal{E}}_{j-1:m} + 3\tilde{\mathcal{E}}_{j:m} \right)^2 \\
 &\quad + \frac{13}{12} \left(\tilde{\mathcal{E}}_{j-2:m} - 2\tilde{\mathcal{E}}_{j-1:m} + \tilde{\mathcal{E}}_{j:m} \right)^2, \\
 IS_m^{+2} &= \frac{1}{4} \left(-\tilde{\mathcal{E}}_{j-1:m} + \tilde{\mathcal{E}}_{j+1:m} \right)^2 \\
 &\quad + \frac{13}{12} \left(\tilde{\mathcal{E}}_{j-1:m} - 2\tilde{\mathcal{E}}_{j:m} + \tilde{\mathcal{E}}_{j+1:m} \right)^2, \\
 IS_m^{+3} &= \frac{1}{4} \left(-3\tilde{\mathcal{E}}_{j:m} + 4\tilde{\mathcal{E}}_{j+1:m} - \tilde{\mathcal{E}}_{j+2:m} \right)^2 \\
 &\quad + \frac{13}{12} \left(\tilde{\mathcal{E}}_{j:m} - 2\tilde{\mathcal{E}}_{j+1:m} + \tilde{\mathcal{E}}_{j+2:m} \right)^2. \tag{15}
 \end{aligned}$$

Here $\epsilon = 10^{-6} \times \{(\xi_x/J)_{j+1/2}^2 + (\xi_y/J)_{j+1/2}^2 + (\xi_z/J)_{j+1/2}^2\}$ is a small number added to prevent division by zero, and p is set to be 2 in this study, where $(\xi_x/J)_{j+1/2} = 1/2\{(\xi_x/J)_j + (\xi_x/J)_{j+1}\}$. The definition of ϵ is modified as the scheme recovers to the original form when the Cartesian equally-spaced grid is adopted.

Finally, the characteristic form of the WENO numerical flux is transformed into its conservative form as follows:

$$E_{j+1/2}^{WENO+} = \sum_m \mathcal{E}_{j+1/2:m}^{WENO+} r_{j:m}, \tag{16}$$

where $r_{j:m}$ denotes the m th right eigenvector of the Roe matrix based on the j and $j+1$ th grid points. Similarly, we can get $E_{j+1/2}^{WENO-}$, and the WENO numerical flux is constructed as follows:

$$E_{j+1/2}^{WENO} = E_{j+1/2}^{WENO+} + E_{j+1/2}^{WENO-}. \tag{17}$$

Finally, the approximation of the derivative is obtained as follows:

$$\frac{\partial E}{\partial \xi} = \frac{1}{\Delta \xi} \left(E_{j+1/2}^{WENO} - E_{j-1/2}^{WENO} \right). \tag{18}$$

In this procedure, we should use the upwinding flux with varying metrics and

the Jacobian for a formal order of accuracy. The use of upwinding flux prevents us from using recent conservative metrics techniques because freestream preservation using conservative metrics requires the use of the unique scheme for the convection terms and the metrics terms in each direction, and it is difficult to use upwinding schemes for the evaluation of the metrics. In addition, as indicated by Nonomura et al.[25], the weights are not ideal because of oscillating metric quantities when the freestream is imposed. The technique proposed in this study resolves these problems.

2.3 A technique on WENO for Freestream Preserving

We divide the WENO flux into the consistent and numerical dissipation parts, as in Jiang.[16] The WENO flux can be rewritten as

$$\begin{aligned}
 \tilde{E}_{j+1/2:m}^{WENO} = & \underbrace{\frac{1}{60} (\tilde{E}_{j-2} - 8\tilde{E}_{j-1} + 37\tilde{E}_j + 37\tilde{E}_{j+1} - 8\tilde{E}_{j+2} + \tilde{E}_{j+3})}_{\text{consistent part}} \\
 & - \underbrace{\sum_m \frac{r_{j:m}}{60} \{ (20w_m^{+,1} - 1) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,1} - 10 ((w_m^{+,1} + w_m^{+,2}) + 5) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,2} + \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,3} \}}_{\text{dissipation part by positive upwinding flux}} \\
 & + \underbrace{\sum_m \frac{r_{j:m}}{60} \{ (20w_m^{-,1} - 1) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,1} - 10 ((w_m^{-,1} + w_m^{-,2}) + 5) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,2} + \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,3} \}}_{\text{dissipation part by negative upwinding flux}},
 \end{aligned} \tag{19}$$

where

$$\begin{aligned}
\tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,1} &= -\tilde{\mathcal{E}}_{j+1}^+ + 3\tilde{\mathcal{E}}_j^+ - 3\tilde{\mathcal{E}}_{j-1}^+ + \tilde{\mathcal{E}}_{j-2}^+ \\
\tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,2} &= -\tilde{\mathcal{E}}_{j+2}^+ + 3\tilde{\mathcal{E}}_{j+1}^+ - 3\tilde{\mathcal{E}}_j^+ + \tilde{\mathcal{E}}_{j-1}^+ \\
\tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,3} &= -\tilde{\mathcal{E}}_{j+3}^+ + 3\tilde{\mathcal{E}}_{j+2}^+ - 3\tilde{\mathcal{E}}_{j+1}^+ + \tilde{\mathcal{E}}_j^+ \\
\tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,1} &= -\tilde{\mathcal{E}}_{j+3}^- + 3\tilde{\mathcal{E}}_{j+2}^- - 3\tilde{\mathcal{E}}_{j+1}^- + \tilde{\mathcal{E}}_j^- \\
\tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,2} &= -\tilde{\mathcal{E}}_{j+2}^- + 3\tilde{\mathcal{E}}_{j+1}^- - 3\tilde{\mathcal{E}}_j^- + \tilde{\mathcal{E}}_{j-1}^- \\
\tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,3} &= -\tilde{\mathcal{E}}_{j+1}^- + 3\tilde{\mathcal{E}}_j^- - 3\tilde{\mathcal{E}}_{j-1}^- + \tilde{\mathcal{E}}_{j-2}^-.
\end{aligned} \tag{20}$$

First, the consistent part is discussed. For this part, we can use the conservative metric technique because it is in the form of a central difference scheme. The error caused by the metric identity is no longer generated if the conservative metric quantity is evaluated by the unique scheme for each direction[2], as shown below.

$$x_{j+1/2} = \frac{1}{60} (x_{j-2} - 8x_{j-1} + 37x_j + 37x_{j+1} - 8x_{j+2} + x_{j+3}) \tag{21}$$

$$\left(\frac{\partial x}{\partial \xi}\right)_j = \frac{1}{\Delta \xi} (x_{j+1/2} - x_{j-1/2}). \tag{22}$$

Next, we focus on the numerical dissipation terms. If the freestream is imposed in these terms, these dissipation terms are not zero because Eqs. 20 are not zero owing to the metric terms. Therefore, we propose a form as discussed below.

The Taylor expansion of the numerical dissipation term with ideal weights is as follows:

$$\underbrace{\sum_m \frac{1}{60} \{ (20w_m^{+,1} - 1) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,1} - 10 ((w_m^{+,1} + w_m^{+,2}) + 5) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,2} + \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,3} \}}_{\text{dissipation part by positive upwinding flux}} = \frac{\partial^5 \tilde{\mathcal{E}}^+}{\partial \xi^5} \Delta \xi^5 + O(\Delta \xi^5). \tag{23}$$

Interestingly, the numerical dissipation term has only high-order derivatives

and purely works as numerical dissipation. Therefore, the upwinding flux in the numerical dissipation part can be modified under the following two conditions: 1) upwinding flux should be smooth in the region where the flow variables are smooth and 2) it works as the appropriate numerical dissipation for various flow fields, including shock waves. We can choose upwinding flux so that the numerical dissipation terms disappear when the freestream is imposed, maintaining the two conditions mentioned above. In this study, we introduce a slight change in metric term treatment for the upwinding flux as follows:

$$\mathcal{E}_{k,m}^{\pm mod} = l_{j;m} \left\{ \left(\frac{\xi_x}{J} \right)_{j+1/2} E_k + \left(\frac{\xi_y}{J} \right)_{j+1/2} F_k + \left(\frac{\xi_z}{J} \right)_{j+1/2} G_k \pm \lambda_m \left(\frac{Q_k}{J_{j+1/2}} \right) \right\}, \quad (24)$$

where

$$\left(\frac{\xi_x}{J} \right)_{j+1/2} = \frac{1}{2} \left\{ \left(\frac{\xi_x}{J} \right)_j + \left(\frac{\xi_x}{J} \right)_{j+1} \right\}. \quad (25)$$

In the equations above, the metric terms are frozen only for the upwinding flux. This technique leads to zero dissipation when the freestream is imposed, and the freestream seems to be preserved. It should be noted that this zero dissipation is also achieved for the original WENO scheme with the Cartesian grid. This numerical flux seems to be smooth if the flow field is smooth, and the formal order of accuracy is expected to be maintained. Also, this formulation recovers the original WENO scheme when the Cartesian grid is used. Additionally, the numerical stability does not seem to change much. To validate these expectations, numerical tests are demonstrated in the next section. The extension of this technique to other flux-splitting methods is trivial. This idea can also be applied to the recent high-resolution WENO schemes.

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In this study, this technique is applied to the state-of-the-art WENO scheme developed by Hu et al. (which is a sixth-order central-and-upwinding-hybrid WENO scheme (WENOCU)), [14] and the effectiveness and generality of the technique for recent high-resolution WENO schemes are discussed.

2.4 Implementations Tested in This Paper and Other Computational Settings

The effectiveness of the present technique is discussed by examining five implementations of WENO, as listed in Table . The first implementation, “WENO,” is the standard WENO scheme.¹ The second implementation, “WENO-CENTFP,” is the WENO scheme with the present technique only for the consistent part, and the third implementation, “WENO-DISFP,” is WENO with the present technique only for the dissipation part. The fourth one, “WENO-FP,” is WENO with the present technique for both consistent and dissipation parts, and the last implementation, “WENOCU-FP,” is the WENO scheme provided by Hu et al. [14] with the present technique for both consistent and dissipation parts. The detailed procedure for WENOCU-FP is presented in Appendix. A. The two implementations, WENO-CENTFP and WENO-DISFP, are computed to isolate the effects of the two different error sources. The elimination of errors by this technique is mainly discussed in the next section. The applicability of this technique to WENOCU will be discussed together. For the numerical tests in the next sections, a third-order total variation diminishing Runge-Kutta scheme [13] is used for time integration.

¹ This implementation is different from the one used in Reference [25]. The definition of ϵ and the flux splitting method are different each other.

In this section, we discuss the results of the numerical tests of freestream, vortex, and shock flows and flow around a delta wing.

3.1 Freestream

The wavy grid for the freestream preservation test, shown in Fig. 1, is expressed as follows:

$$\begin{aligned}
 x_{j,k,l} &= x_{\min} \\
 &+ \Delta x_0 \left[(j-1) + A_x \sin \frac{n_{xy}\pi(k-1)\Delta y_0}{L_y} \sin \frac{n_{xz}\pi(l-1)\Delta z_0}{L_z} \right], \\
 y_{j,k,l} &= y_{\min} \\
 &+ \Delta y_0 \left[(k-1) + A_y \sin \frac{n_{yz}\pi(l-1)\Delta z_0}{L_z} \sin \frac{n_{yx}\pi(j-1)\Delta x_0}{L_x} \right], \\
 z_{j,k,l} &= z_{\min} \\
 &+ \Delta z_0 \left[(l-1) + A_z \sin \frac{n_{zx}\pi(j-1)\Delta x_0}{L_x} \sin \frac{n_{zy}\pi(k-1)\Delta y_0}{L_y} \right], \quad (26)
 \end{aligned}$$

where

$$\begin{aligned}
 j &= 1, 2, \dots, j_{\max}, \\
 k &= 1, 2, \dots, k_{\max}, \\
 l &= 1, 2, \dots, l_{\max}, \\
 \Delta x_0 &= \frac{L_x}{j_{\max} - 1}, \\
 \Delta y_0 &= \frac{L_y}{k_{\max} - 1}, \\
 \Delta z_0 &= \frac{L_z}{l_{\max} - 1}, \\
 x_{\min} &= -\frac{L_x}{2}, \\
 y_{\min} &= -\frac{L_y}{2}, \\
 z_{\min} &= -\frac{L_z}{2}. \quad (27)
 \end{aligned}$$

The wavy grid parameters used in this test are $j_{\max} = k_{\max} = l_{\max} = 21$, $L_x = L_y = L_z = 4$, $A_x = A_y = A_z = 1$, and $n_{xy} = n_{xz} = \dots = 4$. The specific heat ratio of the fluid in the test cases is set to $\gamma = 1.4$, and the fluid is nondimensionalized by the density and the speed of sound for the freestream condition. In this test, an x -direction freestream of Mach number 0.5 is imposed. Thus, the y -direction velocity v and z -direction velocity w are expected to remain machine zero. The boundary conditions are set to the same values as those in the initial condition. The time step Δt is set to 0.1. In this test, double-precision real variables (64 bit) are used in the computational program.

Flow fields are examined after 100 steps. Table 2 shows the L_2 errors of v and w . The results shown in the table demonstrate that both WENO-FP and WENOCU-FP have errors less than 10^{-14} , close to machine zero for double-precision computation. This shows that the technique introduced in this paper works well. On the other hand, the results for WENO show a very large error. This demonstrates that the original WENO scheme does not satisfy the SCL identities on curvilinear grids, as discussed by Nonomura et al.[25]

The errors caused by the consistent and dissipation parts are isolated next. Only the error of the consistent part appears in WENO-DISFPF, and only the error of the dissipation part appears in WENO-CENTFP. Neither implementation preserves the freestream, though the error of WENO-CENTFP is greater than that of WENO-DISFPF. This shows that the error caused by the dissipation part is more significant in the original WENO scheme than that by the consistent part. It implies that special attention should be given to the dissipation part of the WENO scheme when a curvilinear grid is adopted.

Finally, the results of WENOCU-FP show that this implementation can also preserve the freestream. This illustrates that the technique proposed in this paper is also applicable to the WENOCU scheme proposed by Hu et al. and other WENO implementations.

3.2 Moving Vortex

Here two-dimensional moving vortex problems on wavy and randomized grids are computed, and the results are discussed. The specific heat ratio of the fluid in the test cases is set as $\gamma = 1.4$, and the fluid is nondimensionalized by the density and the speed of sound for the freestream condition as in the freestream preservation test. An isentropic vortex centered at $(x_c, y_c) = (0, 0)$ is created in the Mach 0.5 freestream as the initial condition. The velocity, temperature, and entropy fluctuations of the isentropic vortex are expressed as follows:

$$\begin{aligned}(\delta u, \delta v) &= \varepsilon \tau e^{\alpha(1-\tau^2)} (\sin \theta, -\cos \theta), \\ \delta T &= -\frac{(\gamma - 1)\varepsilon^2}{4\alpha\gamma} e^{2\alpha(1-\tau^2)}, \\ \delta S &= 0,\end{aligned}\tag{28}$$

where $\tau = r/r_c$ and $r = [(x - x_c)^2 + (y - y_c)^2]^{1/2}$. Here $r_c = 1.0$ denotes the vortex core length, $\alpha = 0.204$ denotes the parameter of the length scale of vortex decay, and $\varepsilon = 0.02$ denotes the vortex strength. $T = p/\rho$ is the temperature, and $S = p/\rho^\gamma$ is the entropy. The boundary conditions for the ξ - and η -directions are set to be periodic. The computation is performed until time $t = 40$, at which point the vortex moves back to the original position of the initial condition in the grid system defined later. The exact solution of this problem preserves the vortex strength.

The vortex flow on wavy grids is tested. A two-dimensional wavy grid is formulated as follows:

$$\begin{aligned}
 x_{j,k,l} &= x_{\min} \\
 &+ \Delta x_0 \left[(j-1) + A_x \sin \frac{n_{xy}\pi(k-1)\Delta y_0}{L_y} \right], \\
 y_{j,k,l} &= y_{\min} \\
 &+ \Delta y_0 \left[(k-1) + A_y \sin \frac{n_{yx}\pi(j-1)\Delta x_0}{L_x} \right],
 \end{aligned} \tag{29}$$

where

$$\begin{aligned}
 j &= 1, 2, \dots, j_{\max}, \\
 k &= 1, 2, \dots, k_{\max}, \\
 \Delta x_0 &= \frac{L_x}{j_{\max} - 1}, \\
 \Delta y_0 &= \frac{L_y}{k_{\max} - 1}, \\
 x_{\min} &= -\frac{L_x}{2}, \\
 y_{\min} &= -\frac{L_y}{2}.
 \end{aligned} \tag{30}$$

The parameters for the wavy grid are $L_x = L_y = 20$, $A_x/\Delta x_0 = 0.6$, $A_y/\Delta y_0 = 0.6$, and $n_{xy} = n_{yx} = 4$. These parameters generate severe wavy grids. Three different grids, 21×21 , 41×41 , and 81×81 , are used. Here Δt are set to be 0.25, 0.0625, and 0.0015625 for the 21×21 , 41×41 , and 81×81 grids, respectively. These time steps are carefully selected so as to ignore the error in third-order time integration, as discussed later regarding the order of spatial accuracy.

The results on the coarsest grid are used to discuss the general characteristics of the implementations as follows. Figures 2 and 3 show the contours of the vorticities and swirl velocities for the coarsest grid at $t = 40$ for the five

implementations of WENO. These figures demonstrate that WENO-FP can resolve the vortex, whereas WENO cannot because of significant errors from the grid. This illustrates that the technique works well for vortex problems on the wavy grids.

Next, the errors caused in the consistent and dissipation parts are isolated. Figures 2 and 3 show that the WENO-DISFPF scheme, which only includes the error in the consistent part, can resolve the vortex better, whereas WENO-CENTFP, which only includes the error in dissipation part, yields a result similar to the poor result of WENO. This again shows that the error in the dissipation part is significant and should be more carefully considered than that in the consistent part.

The results of the WENOCU-FP scheme show that it can resolve the vortex without significant errors in wavy grids, and the resolution of the vortex is much improved compared with that of WENO-FP. This shows that the technique developed here can be used in the WENOCU scheme without the loss of its high resolution.

Finally, we examine the grid convergence rate for all the implementations. Figure 4 shows the L2 error of v component for various grid density. In Fig. 4, the results using a uniform grid of the same size are plotted as the reference. The grid convergence rates are shown in Table 3. This table shows that results with the wavy grid have larger error than with the uniform grid, and the grid convergence rate becomes slightly smaller with a wavy grid than with a uniform grid if we use appropriate formulation (This is inversed for inappropriate formulation (e.g. WENO) on a coarser grid). The latter implies that the formal order of accuracy is slightly degraded by the curvilinear grid

Next, the vortex preserving property on randomized grids is examined. Uniform grids constructed for the region $-10 < x < 10$ and $-10 < y < 10$ with 21×21 grid points are randomized with 20% magnitude grid spacing in a random direction. Here Δt is set to 0.25. Figure 5(a) shows the computational grid and vorticity magnitude distribution for the initial condition corresponding to the exact solution at $t = 40$. This problem is severer than the previous wavy grid problem.

The corresponding results are shown in Figs. 5 and 6. The results for WENO, WENO-CENTFP, and WENO-DISPFP have large numerical errors. This indicates that errors both from the consistent part and the dissipation part are significant, whereas the results of WENO-DISPFP have relatively small errors compared with those of WENO-CENTFP (see Table 5). The trend that the error in the dissipation part is more significant does not change. On the other hand, the results of WENO-FP and WENOCU-FP show that the vortices are well captured by these two implementations. This illustrates that eliminating both errors in the consistent and in the dissipation parts is important when a grid with severe distortion is adopted. The vortex is better captured by WENOCU-FP than by WENO-FP, and this example again shows that the superiority of WENOCU to WENO in resolution does not change even with the technique proposed in this study.

The supersonic flow past a cylinder[17] is solved on a randomized curvilinear grid, which seems to be a severe condition from the vortex problem as previously discussed. The grid system is given as follows:

$$\begin{aligned}
 x &= (R_x - (R_x - 1)\eta') \cos(\theta(2\xi' - 1)) \\
 y &= (R_y - (R_y - 1)\eta') \sin(\theta(2\xi' - 1)) \\
 \xi' &= \frac{\xi - 1}{j_{\max} - 1} \\
 \eta' &= \frac{\eta - 1}{k_{\max} - 1} \\
 \xi &= j + \text{Random}_j \\
 \eta &= k + \text{Rondom}_k \\
 \sqrt{\text{Random}_j^2 + \text{Romdom}_k^2} &= \begin{cases} 0.2 & \text{inner points} \\ 0 & \text{boundary points} \end{cases} \\
 j &= 1, 2, \dots, j_{\max}, \\
 k &= 1, 2, \dots, k_{\max},
 \end{aligned} \tag{31}$$

With the parameters as follows: $j_{\max} = 81$, $k_{\max} = 61$, $R_x = 3$, and $R_y = 6$. Thus, the grid number is 81×61 . At the $k = 1$ boundary, a slip wall condition is imposed. With this grid, a freestream of Mach 2.0 is imposed as the initial condition, and the steady state of the flow fields are discussed. The time step $\Delta t = 0.005$ is used and 5,000 steps are computed, at which point the solution converges. The grid is shown in Fig. 7(a).

Results are shown in Fig. 7. The results for WENO and WENO-CENTFP have significant errors, whereas those for the other schemes do not. The results illustrate that the technique presented in this paper eliminates the error and does not degrade the shock-capturing ability of WENO and WENOCU.

ing problem on the randomized grid: WENO-DISPFP can solve the flow field without significant error despite the error from the consistent part, which is significant in the test case of the vortex on the randomized grid. Again, this shows that the error from the dissipation part is more significant than that from the consistent part.

3.4 Double Mach Reflection

The double Mach reflection problem[36] is solved on a randomized grid. The initial condition is defined as follows:

$$\begin{aligned}
 U(x, y, 0) &= \begin{cases} U_L & \text{if } x - y \tan \frac{\pi}{6} \leq 1/6 \\ U_R & \text{if } x - y \tan \frac{\pi}{6} > 1/6 \end{cases} \\
 U_L &= (\rho_L, u_L, v_L, p_L)^T = (8.0, 7.145, -4.125, 116.5)^T \\
 U_R &= (\rho_R, u_R, v_R, p_R)^T = (1.4, 0, 0, 1)^T \\
 &0 \leq x \leq 4, \quad 0 \leq y \leq 1
 \end{aligned} \tag{32}$$

The grid number is set to 241×61 , and the CFL number is set to 0.6. The computation is conducted till $t = 0.2$. The grid is random for all inner points, but not for the boundary points. The randomness of the inner grid points is limited to 2% of the grid spacing because some implementations suffer from the carbuncle phenomenon with more severe randomness. Although a mild randomness quantity of 2% is adopted, the errors in the inappropriate implementations are significant as shown later. The grid is shown in Fig. 8(a).

The results shown in Fig. 8 illustrate a similar trend as that in the previ-

ous steady supersonic flow problem. They show that the technique presented in this paper eliminates the error and does not degrade the shock-capturing ability of WENO and WENOCU even for the unsteady problem. Also, WENO-DISPFP can solve the flow field without significant errors, whereas WENO-CENTFP has significant errors. This further illustrates that the error from the dissipation part is more significant than that from the consistent part.

Finally, comparing the WENO-FP and WENOCU-FP schemes, the resolution of a shear layer from the triple point by WENOCU-FP is much higher than that by WENO-FP. The results show that the higher resolution of WENOCU is maintained even using the technique proposed in this paper.

3.5 *Practical Application: Flow around a Delta Wing*

Finally, the proposed technique is applied to the practical application of flow around a delta wing. Navier-Stokes equations nondimensionalized by the root chord length, the density, and the speed of sound for the freestream condition, are solved. The freestream Mach number, the Reynolds number based on the root chord length, and angle of attack are set to be 0.3, 10^6 , and 20.5[deg], respectively. The delta wing has a 76[deg] sweep angle and a sharp leading edge. These computational conditions are similar to those of an experimental study[15].

The second-order central difference scheme is employed, while the Baldwin-Lomax turbulence model[5] with Degani and Schiff modification[8] is adopted together for the viscous terms in Navier-Stokes equations. The lower-upper alternative direction implicit factorization scheme[12] is employed for the time

integration. The number of computational grid shown in Fig. 9 are set to be $55 \times 65 \times 35$, and time step Δt is set to be 0.01. Approximately 4000 steps are computed, which is sufficient for converging the flow fields with a three-order reduction in the residual for the converged case. We examine whether the computation is robust or not on the practical highly-skewed grid.

In this test case, only WENO, WENO-FP, and WENOCU-FP are tested because the discussion on the two different sources of error appear to be sufficient with the previous examples. As expected, computation with WENO scheme blows up at the 1712 step, while computation with WENO-FP scheme successfully finishes. This seems to be because the errors in WENO making the computation unstable is properly eliminated by WENO-FP. On the other hand, unfortunately, the computation with WENOCU-FP also blows up at the 611 step even the errors seem to be eliminated as in the previous problems (This is confirmed by the computation with the freestream boundary condition on all the boundaries of the delta wing grid.). This implies that the grid used in this problem is further skewed around the leading edge than previous problems, and WENOCU-FP blows up due to a highly skewed grid because the WENOCU seems to have high resolution but less robustness in our experience, though such a less robustness of WENOCU was not reported in the original paper.[14] We recommend that WENOCU-FP on a curvilinear grid system should be used with a smoother grid, even if we use the technique presented in this paper. The converged flow fields or ones before blow up are shown in Figs. 10 and 11 the blow up time steps for the scheme tested are summarized in Table 6. The results show that the WENO-FP results agree well with the experimental data.

A technique for a finite-difference WENO to preserve the freestream on a curvilinear grid is introduced. The technique divides the finite-difference WENO into two parts: 1) a consistent central difference part and 2) a numerical dissipation part. For the consistent central difference part, the conservative metric technique is directly adopted. For the numerical dissipation part, it is proposed that the metric term is frozen for constructing the upwinding flux. This treatment only affects the numerical dissipation part, and the order of accuracy is interestingly maintained. With this technique, the freestream is perfectly preserved, and the resolution of vortices is also much improved on wavy and random grids. This technique also works well with high-resolution versions of WENO, while maintaining their high resolution. In addition, test cases with shock waves illustrate that this technique does not degrade the shock-capturing capability of WENO schemes. Finally, the present authors believe that this technique can be straightforwardly extended to WENO scheme on the moving and deforming grid though it is left for a future work.

Appendix. A formulation of WENOCU

In this appendix, the formulation of WENOCU is introduced, and the split form of WENOCU is derived. First, the upwinding fluxes are constructed on the grid point as follows:

$$\mathcal{E}_{k,m}^{\pm} = l_{j;m} \left(\frac{\xi_{x_k} E_k + \xi_{y_k} F_k + \xi_{z_k} G_k \pm \lambda_m Q}{J_k} \right) \quad (k = j - 2, j - 1, j, j + 1, j + 2, j + 3). \quad (33)$$

Then the characteristic form of the WENO numerical flux $\tilde{\mathcal{E}}_{j+1/2,m}^{WENO+}$ is constructed as follows:

$$\tilde{\mathcal{E}}_{j+1/2,m}^{WENOCU+} = w_m^1 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+1} + w_m^2 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+2} + w_m^3 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+3} + w_m^4 \tilde{\mathcal{E}}_{j+1/2,m}^{WENO+4}, \quad (34)$$

where

$$\begin{aligned} \mathcal{E}_{j+1/2,m}^{WENO+1} &= \frac{1}{3} \mathcal{E}_{j-2:m}^+ - \frac{7}{6} \mathcal{E}_{j-1:m}^+ + \frac{11}{6} \mathcal{E}_{j:m}^+, \\ \mathcal{E}_{j+1/2,m}^{WENO+2} &= -\frac{1}{6} \mathcal{E}_{j-1:m}^+ + \frac{5}{6} \mathcal{E}_{j:m}^+ + \frac{1}{3} \mathcal{E}_{j+1:m}^+, \\ \mathcal{E}_{j+1/2,m}^{WENO+3} &= \frac{1}{3} \mathcal{E}_{j:m}^+ + \frac{5}{6} \mathcal{E}_{j+1:m}^+ - \frac{1}{6} \mathcal{E}_{j+2:m}^+, \\ \mathcal{E}_{j+1/2,m}^{WENO+4} &= \frac{11}{6} \mathcal{E}_{j+1:m}^+ - \frac{7}{6} \mathcal{E}_{j+2:m}^+ + \frac{1}{3} \mathcal{E}_{j+3:m}^+, \end{aligned} \quad (35)$$

and w^1 , w^2 , w^3 and w^4 are nonlinear weights. These nonlinear weights are determined as follows:

$$w_m^{+k} = \frac{\alpha_{j:m}^{+k}}{\alpha_{j:m}^{+1} + \alpha_{j:m}^{+2} + \alpha_{j:m}^{+3}} (k = 1, 2, 3), \quad (36)$$

where

$$\alpha_{j:m}^{+k} = d^{+k} \left(C + \frac{\tau_6}{\beta_{j:m}^k + \epsilon} \right) \quad (k = 1, 2, 3). \quad (37)$$

Here, the ideal weights and smooth indicators are as follows²:

² The coefficients in the original paper[14] has mistake and correct coefficients are shown here.

$$\begin{aligned}
 d^{+1} &= \frac{1}{20}, \quad d^{+2} = \frac{9}{20}, \quad d^{+3} = \frac{9}{20}, \quad d^{+4} = \frac{1}{20}, \\
 C &= 20 \\
 \beta_m^{+1} &= \frac{1}{4} \left(\tilde{\mathcal{E}}_{j-2:m} - 4\tilde{\mathcal{E}}_{j-1:m} + 3\tilde{\mathcal{E}}_{j:m} \right)^2 \\
 &\quad + \frac{13}{12} \left(\tilde{\mathcal{E}}_{j-2:m} - 2\tilde{\mathcal{E}}_{j-1:m} + \tilde{\mathcal{E}}_{j:m} \right)^2, \\
 \beta_m^{+2} &= \frac{1}{4} \left(-\tilde{\mathcal{E}}_{j-1:m} + \tilde{\mathcal{E}}_{j+1:m} \right)^2 \\
 &\quad + \frac{13}{12} \left(\tilde{\mathcal{E}}_{j-1:m} - 2\tilde{\mathcal{E}}_{j:m} + \tilde{\mathcal{E}}_{j+1:m} \right)^2, \\
 \beta_m^{+3} &= \frac{1}{4} \left(-3\tilde{\mathcal{E}}_{j:m} + 4\tilde{\mathcal{E}}_{j+1:m} - \tilde{\mathcal{E}}_{j+2:m} \right)^2 \\
 &\quad + \frac{13}{12} \left(\tilde{\mathcal{E}}_{j:m} - 2\tilde{\mathcal{E}}_{j+1:m} + \tilde{\mathcal{E}}_{j+2:m} \right)^2, \\
 \beta_m^{+4} &= \beta_m^{+6} \\
 &= \frac{1}{120960} [\mathcal{E}_{j-2:m} (271779\tilde{\mathcal{E}}_{j-2:m} - 2380800\tilde{\mathcal{E}}_{j-1:m} + 4086352\tilde{\mathcal{E}}_{j:m} - 3462252\tilde{\mathcal{E}}_{j+1:m} \\
 &\quad + 1458762\tilde{\mathcal{E}}_{j+2:m} - 245620\tilde{\mathcal{E}}_{j+3:m}) \\
 &\quad + \mathcal{E}_{j-1:m} (5653317\tilde{\mathcal{E}}_{j-1:m} - 20427884\tilde{\mathcal{E}}_{j:m} + 17905032\tilde{\mathcal{E}}_{j+1:m} \\
 &\quad - 7727988\tilde{\mathcal{E}}_{j+2:m} + 1325006\tilde{\mathcal{E}}_{j+3:m}) \\
 &\quad + \mathcal{E}_{j:m} (19510972\tilde{\mathcal{E}}_{j:m} - 35817664\tilde{\mathcal{E}}_{j+1:m} + 15929912\tilde{\mathcal{E}}_{j+2:m} - 2792660\tilde{\mathcal{E}}_{j+3:m}) \\
 &\quad + \mathcal{E}_{j+1:m} (17195652\tilde{\mathcal{E}}_{j+1:m} - 15880404\tilde{\mathcal{E}}_{j+2:m} + 2863984\tilde{\mathcal{E}}_{j+3:m}) \\
 &\quad + \mathcal{E}_{j+2:m} (3824847\tilde{\mathcal{E}}_{j+2:m} - 1429976\tilde{\mathcal{E}}_{j+3:m}) \\
 &\quad + \mathcal{E}_{j+3:m} (139633\tilde{\mathcal{E}}_{j+3:m})] \\
 \tau_6 &= \beta_6 - \frac{1}{6} (\beta_1 + 4\beta_2 + \beta_3). \tag{38}
 \end{aligned}$$

Here $\epsilon = 10^{-40} \times \{(\xi_x/J)_{j+1/2}^2 + (\xi_y/J)_{j+1/2}^2 + (\xi_z/J)_{j+1/2}^2\}$ is a very small number added to prevent division by zero. Finally, the numerical flux is constructed similarly to Eq. 16 of the standard WENO scheme.

The modified WENOCU using the present technique is shown as follows:

$$\begin{aligned}
\tilde{E}_{j+1/2:m}^{WENOCU} = & \underbrace{\frac{1}{60} (\tilde{E}_{j-2} - 8\tilde{E}_{j-1} + 37\tilde{E}_j + 37\tilde{E}_{j+1} - 8\tilde{E}_{j+2} + \tilde{E}_{j+3})}_{\text{consistent part}} \\
& - \underbrace{\sum_m \frac{r_{j:m}}{60} \{ (20w_m^{+,1} - 1) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,1} + (10(w_m^{+,3} + w_m^{+,4}) - 5) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,2} + (1 - 20w_m^{+,4}) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime+,3} \}}_{\text{dissipation part by positive upwinding flux}} \\
& + \underbrace{\sum_m \frac{r_{j:m}}{60} \{ (20w_m^{-,1} - 1) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,1} + (10(w_m^{-,3} + w_m^{-,4}) - 5) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,2} + (1 - 20w_m^{-,4}) \tilde{\mathcal{E}}_{j:m}^{\prime\prime\prime-,3} \}}_{\text{dissipation part by negative upwinding flux}}.
\end{aligned} \tag{39}$$

With this form, the conservative metric technique for the consistent part and the frozen metric technique for the dissipation part can be used. The resulting scheme can preserve the freestream as indicated in this paper.

Appendix. B Comparison of Resolution and Efficiency of WENO-FP and WCNS

In this appendix, the resolution and efficiency of WENO-FP and WCNS[25] are briefly compared. Here, fifth-order explicit WCNS[11,22] with Roe's FDS[28] is employed as the representative of similar schemes, WCNS, HWCNS, AWENO, PFWENO and robust WCNS[23]. The more detailed comparison of WENO-FP and the WCNS type scheme are left for the future research, because we should address the many difference in the schemes, e.g. spectral property, flux function and flux splitting, or smooth indicator, fair comparison which is out of scope of the present paper. The comparison here is limited, but we try fair comparisons as possible in the current state. The freestream preservation are satisfied for both implementations and results are omitted. The vortex moving problems used in the present paper are employed for comparison. First, the results for uniform grids are discussed. The contours of the vorticities and plot of swirl velocities for WENO-FP and WCNS are shown in Figs. 13 and 14. As

discussed in the previous study[25] the resolution of WCNS is slightly higher than that of WENO-FP. Then, the results for wavy grids shown in Figs. 15 and 16 show that the trends on wavy grids are similar to those for uniform grid: the resolution of WCNS is slightly higher than that of WENO. This indicates that the WENO-FP works as well as WCNS. On the other hand, the results for a random grid shown in Figs. 17 and 18 show that the error is slightly more suppressed in the results of WCNS than WENO-FP, and the swirl velocity is slightly better captured by WCNS than WENO-FP. These limited results imply that WCNS is better for a highly skewed grid while WENO-FP and WCNS give us almost similar results for a relatively smooth grid.

Then, the computational costs are summarized in the Table 7. The computational cost (time) is measured for the freestream preservation problem adopted in this paper. The obtained computational costs should be just the reference value because the computational code used here is not fully tuned. The freestream preservation technique for WCNS is just in the evaluation of the metrics, and additional cost is negligible. Thus, the costs for WCNS does not change with/without the freestream preservation technique. On the other hand, the present technique for the WENO scheme requires additional procedures compared with the original WENO scheme, and thus computation costs for both schemes are shown. Here, we use Xeon 2.8GHz 8 core processors with auto parallelization. Table 7 shows that the computational costs for WCNS is less than WENO. Here, WCNS also requires the characteristic interpolation of flow variables[23,26] similar to the characteristic interpolation of flux in WENO scheme, which is most expensive in the scheme. The lower computational of WCNS might be because of the more efficient form of smooth indicators in WCNS[23]. The WENO-FP requires additional small

costs by introducing the split form, comparing WENO. From the viewpoint of the computational cost, WCNS seems to be better in the limited comparison in this appendix.

Acknowledgment

This work was supported by KAKENHI (24656522) and (25709009), Strategic Programs for Innovative Research (SPIRE) of High-Performance Computing Initiative (HPCI).

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Summary of the implementations tested in this paper.

		nonlinear weights	conservative metrics	frozen metrics
		for the consistent part		for the dissipation part
WENO	WENO[17]	No	No	No
WENO-CENTFP	WENO[17]	Yes	No	No
WENO-DISFPF	WENO[17]	No	Yes	No
WENO-FP	WENO[17]	Yes	Yes	Yes
WENOCU-FP	WENOCU[14]	Yes	Yes	Yes

L_2 errors of v and w components in the freestream preservation test on a wavy grid.

	v -component	w -component
WENO	8.52×10^{-3}	8.33×10^{-3}
WENO-CENTFP	1.03×10^{-2}	1.02×10^{-2}
WENO-DISPFP	2.13×10^{-3}	2.13×10^{-3}
WENO-FP	1.48×10^{-16}	1.48×10^{-16}
WENOCU-FP	1.42×10^{-15}	1.25×10^{-15}

L_2 errors and convergence rate for v component in the moving vortex on a wavy grid.

wavy grid	grid size	error	order of accuracy
WENO	21×21	3.51×10^{-2}	-
	41×41	1.84×10^{-3}	4.25
	81×81	1.13×10^{-4}	4.02
WENO-CENTFP	21×21	3.37×10^{-2}	-
	41×41	9.07×10^{-3}	4.19
	81×81	6.58×10^{-4}	4.02
WENO-DISFPF	21×21	2.75×10^{-3}	-
	41×41	9.07×10^{-4}	1.60
	81×81	6.58×10^{-5}	3.78
WENO-FP	21×21	2.71×10^{-3}	-
	41×41	9.06×10^{-4}	1.65
	81×81	6.58×10^{-5}	3.78
WENOCU-FP	21×21	2.19×10^{-3}	-
	41×41	1.65×10^{-4}	3.73
	81×81	6.66×10^{-6}	4.63

L_2 errors and convergence rate for v component in the moving vortex on a uniform grid.

uniform grid	grid size	error	order of accuracy
WENO	21×21	1.76×10^{-3}	-
	41×41	2.39×10^{-4}	2.88
	81×81	1.05×10^{-6}	4.51
WENO-CENTFP	21×21	1.76×10^{-3}	-
	41×41	2.39×10^{-4}	2.88
	81×81	1.05×10^{-6}	4.51
WENO-DISFPF	21×21	1.76×10^{-3}	-
	41×41	2.39×10^{-4}	2.88
	81×81	1.05×10^{-6}	4.51
WENO-FP	21×21	1.76×10^{-3}	-
	41×41	2.39×10^{-4}	2.88
	81×81	1.05×10^{-6}	4.51
WENOCU-FP	21×21	2.96×10^{-4}	-
	41×41	1.62×10^{-5}	4.36
	81×81	1.46×10^{-6}	5.53

L_2 errors of v components in the moving vortex on a randomized grid.

	error
WENO	1.71×10^{-1}
WENO-CENTFP	1.54×10^{-1}
WENO-DISFPF	5.21×10^{-2}
WENO-FP	1.87×10^{-3}
WENOCU-FP	1.92×10^{-3}

Blow up time step for the computation of flow around the delta wing.

Blow up time step	
WENO	1712
WENO-FP	No blow up
WENOCU-FP	611

Computational cost (time) for the computation of freestream on the wavy grid.

	computational time
WENO	27.3sec
WENO-FP	27.7sec
WCNS	19.1sec



Fig. 1. A wavy grid for the freestream preservation test.

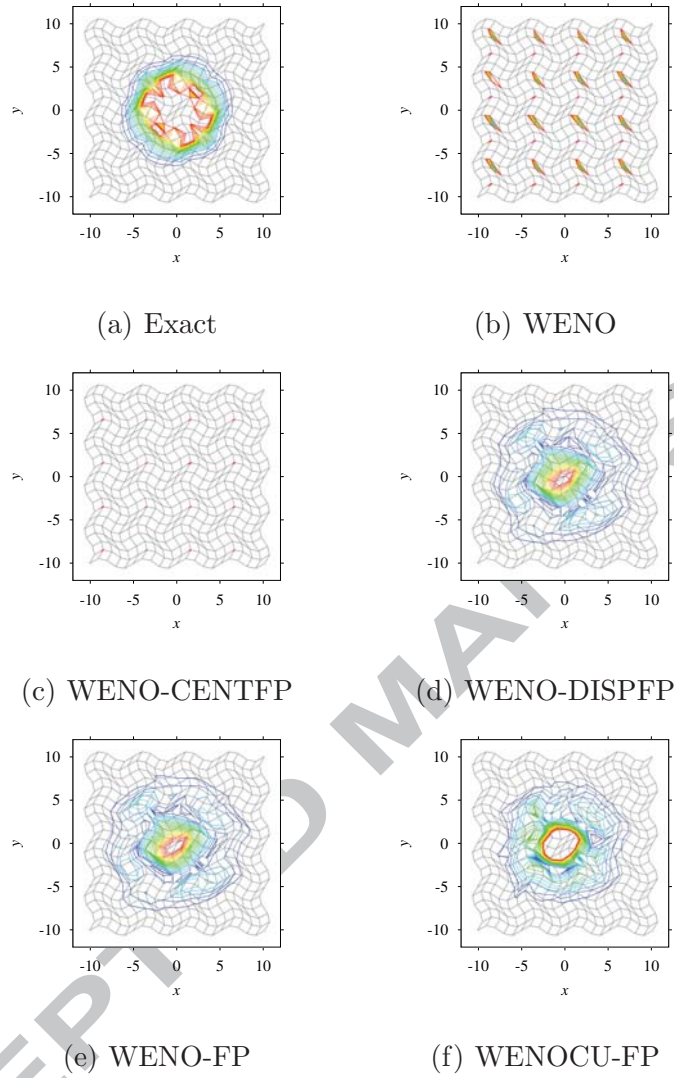


Fig. 2. Vorticity magnitude distributions of the results of the two-dimensional moving vortex problem on the coarsest wavy grid. Here 21 contours from 0.0 to 0.005 are shown.

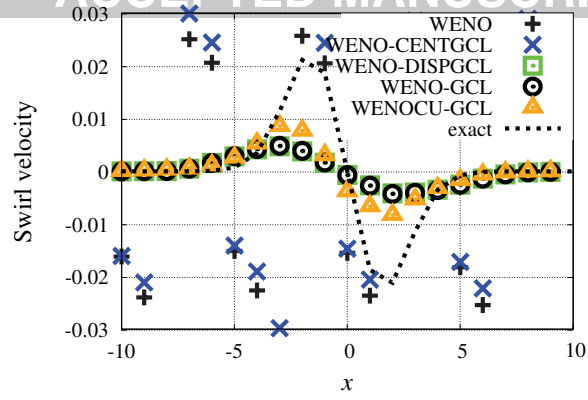
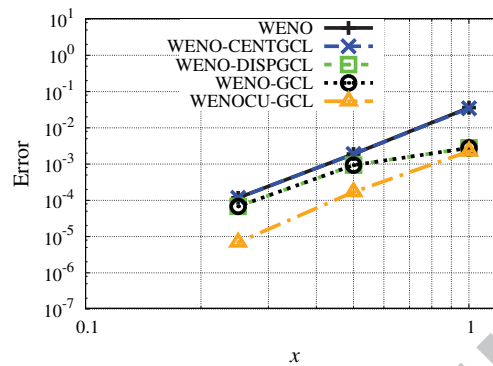
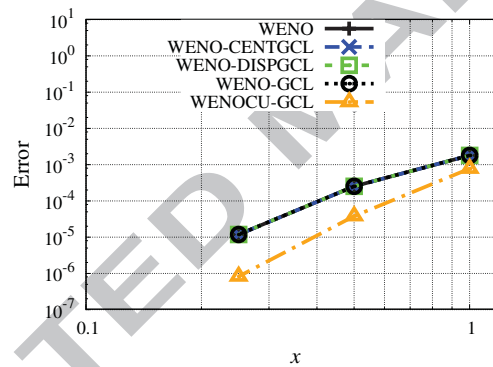


Fig. 3. Swirl velocity of the moving vortex solved on the wavy grid.



(a) Errors on the wavy grids



(b) Errors on a uniform grid (reference)

Fig. 4. Errors of the moving vortex problem with a different grid density.

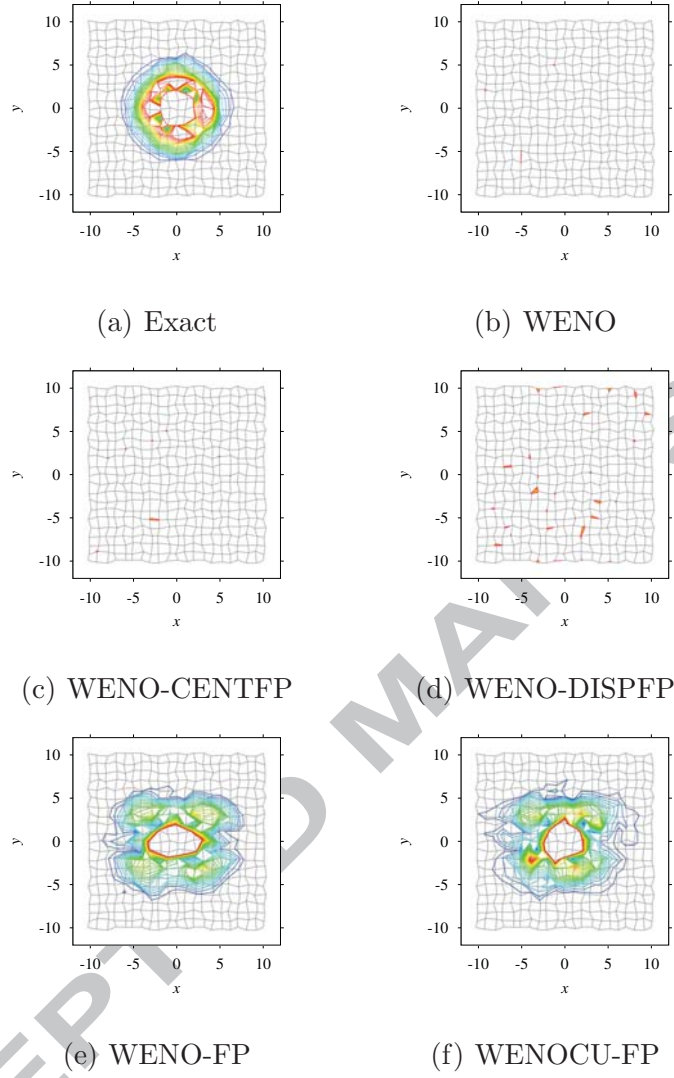


Fig. 5. Vorticity magnitude distributions of the results of the two-dimensional moving vortex problem on a randomized grid. Here, 21 contours from 0.0 to 0.005 are shown.

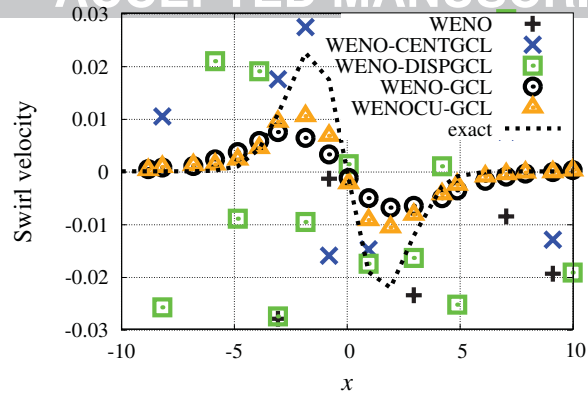


Fig. 6. Swirl velocity of the moving vortex solved on a randomized grid.

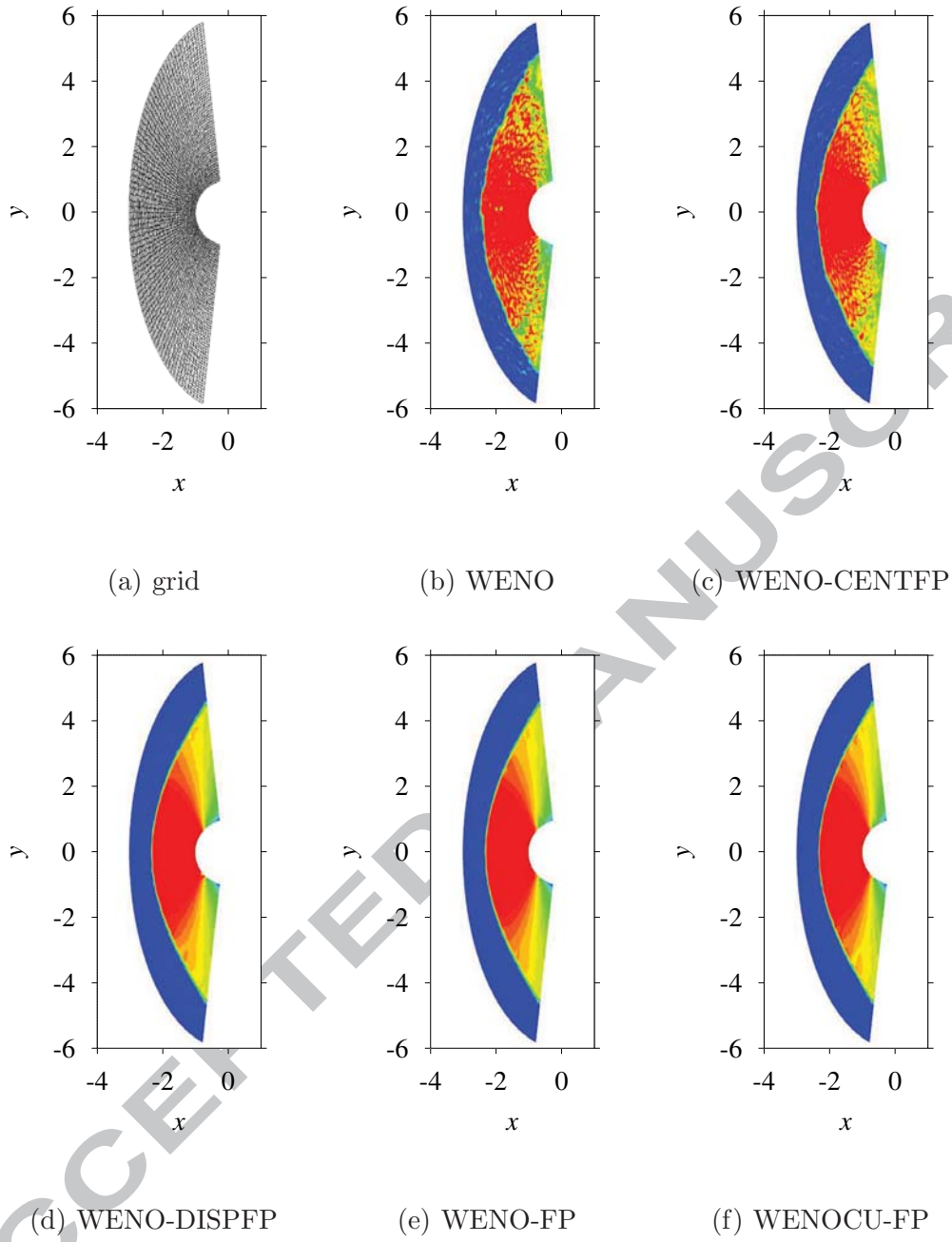


Fig. 7. The pressure distributions of the supersonic flow past a cylinder problem.

Here 21 Contours are drawn from $1/\gamma$ to $4/\gamma$.

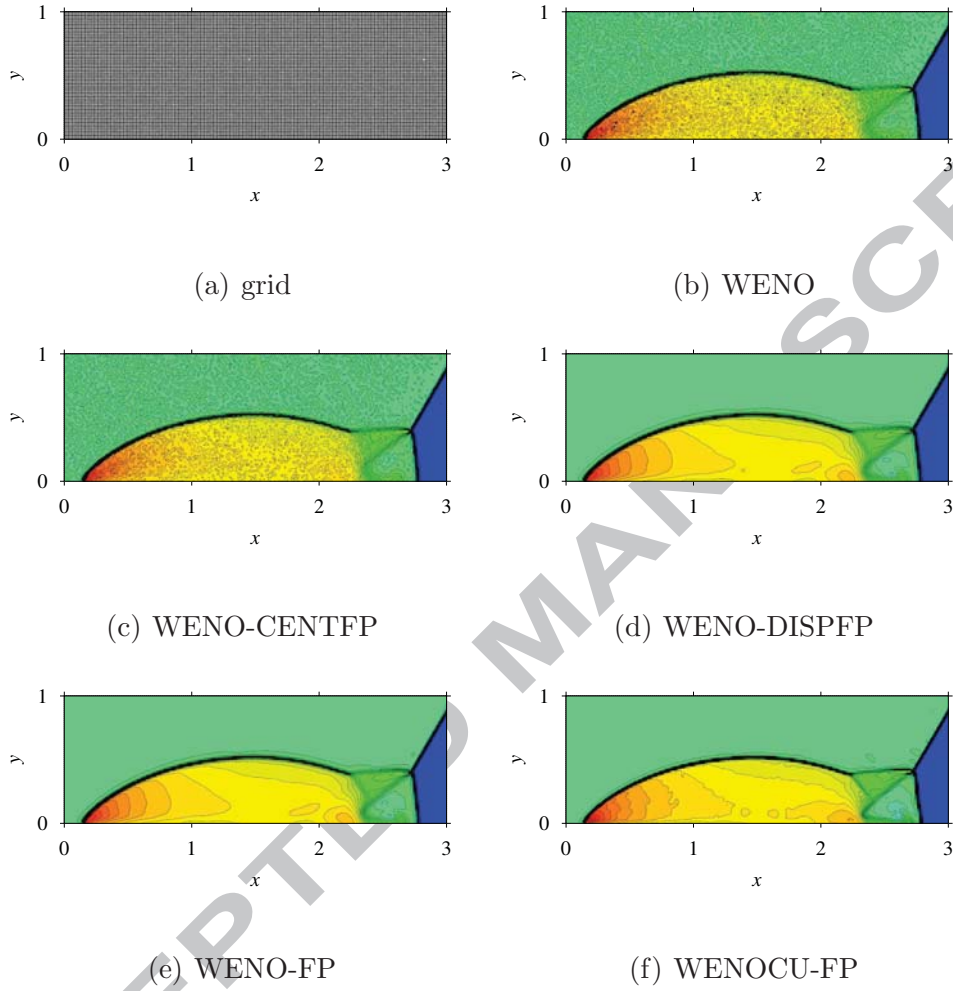


Fig. 8. Density contours of the double Mach reflection problem. Here 40 contours are drawn from 1.88783 to 20.9144.

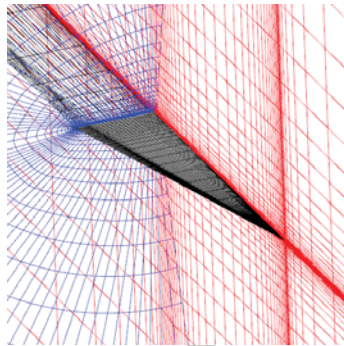
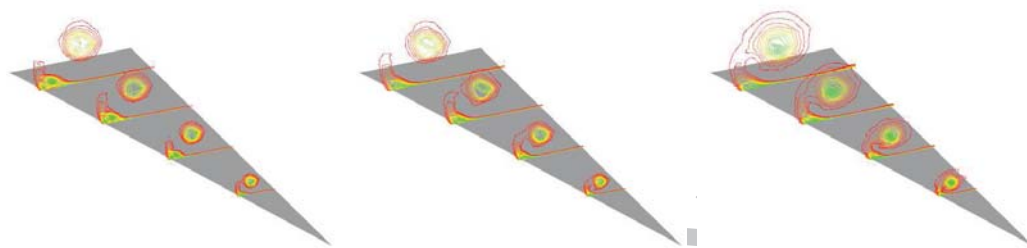


Fig. 9. Grid system for the computation of flow around the delta wing.

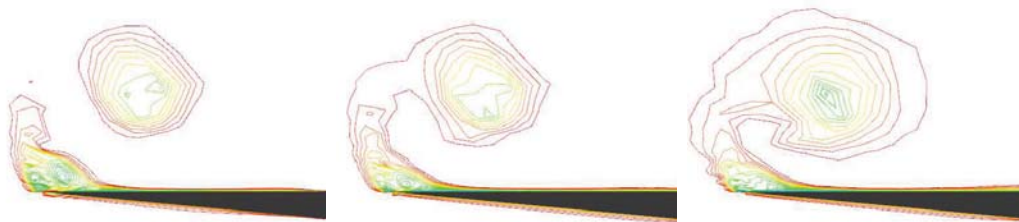


(a) WENO (before blow up)

(b) WENO-FP

(c) WENOCU-FP (before blow up)

Fig. 10. Total pressure distributions of a delta wing. Here, the solutions converged or just before blow up are visualized. The contour range for normalized total pressure is set to be 0.86 to 0.99.



(a) WENO (before blow up)

(b) WENO-FP

(c) WENOCU-FP (before blow up)

Fig. 11. Total pressure distributions of 50% chord length position of a delta wing. Here, the solutions converged or just before blow up are visualized. The contour range for normalized total pressure is set to be 0.86 to 0.99.

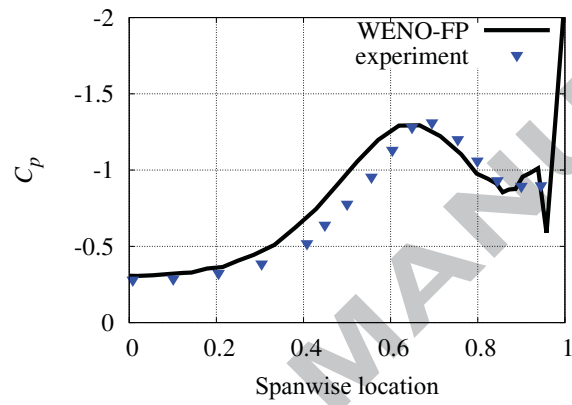


Fig. 12. C_p distribution 50% chord length position of the delta wing computed by WENO-FP.

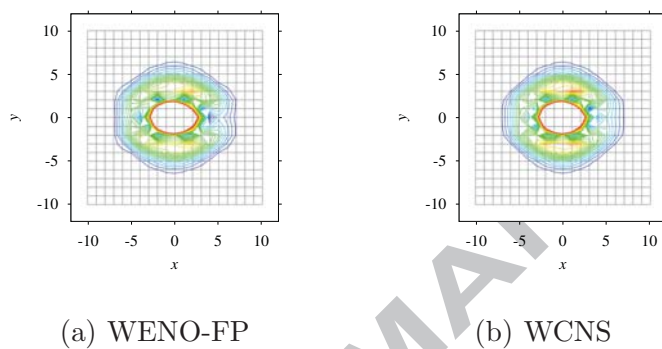


Fig. 13. Vorticity magnitude distributions of WENO-FP and WCNS of the two-dimensional moving vortex problem on the coarsest wavy grid. Here 21 contours from 0.0 to 0.005 are shown.

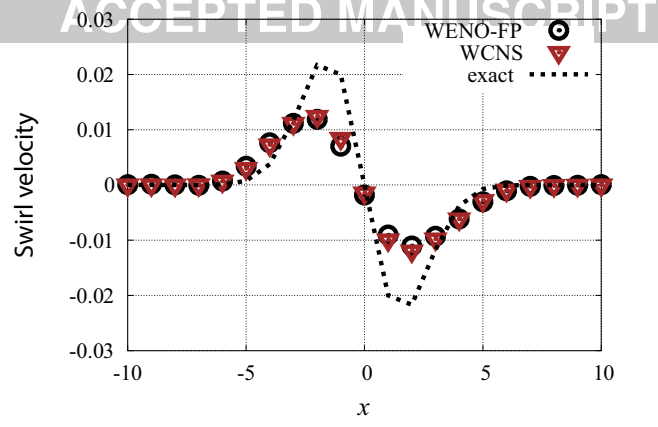


Fig. 14. Swirl velocity of the moving vortex solved by WENO-FP and WCNS the coarsest uniform grid.

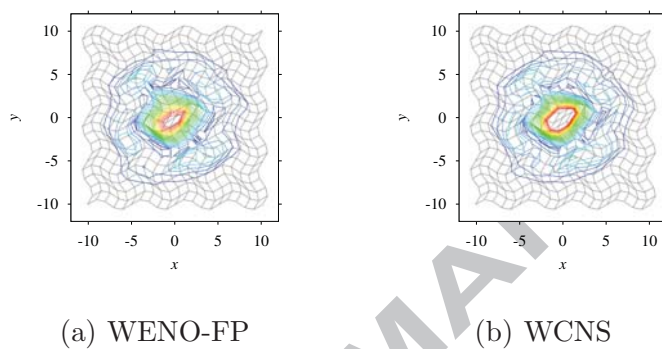


Fig. 15. Vorticity magnitude distributions of WENO-FP and WCNS of the two-dimensional moving vortex problem on the coarsest wavy grid. Here 21 contours from 0.0 to 0.005 are shown.

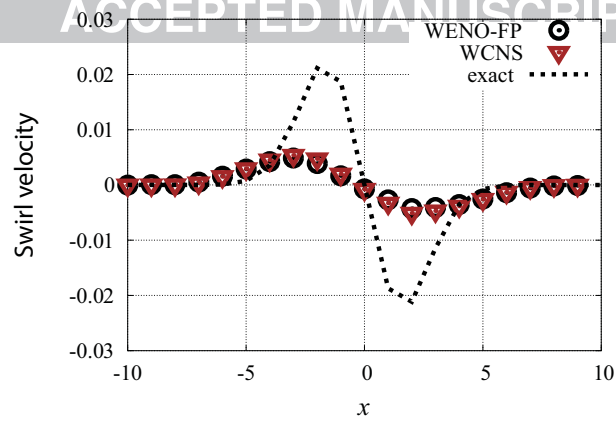


Fig. 16. Swirl velocity of the moving vortex solved by WENO-FP and WCNS on the coarsest wavy grid.

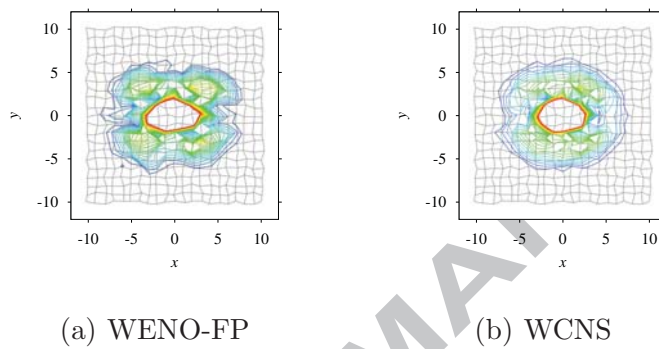


Fig. 17. Vorticity magnitude distributions of WENO-FP and WCNS of the two-dimensional moving vortex problem on the randomized grid. Here 21 contours from 0.0 to 0.005 are shown.

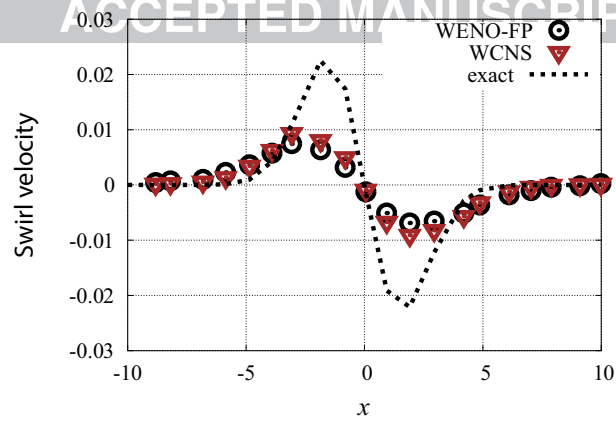


Fig. 18. Swirl velocity of the moving vortex solved by WENO-FP and WCNS on a randomized grid.

- A freestream preservation technique for a standard FDWENO scheme on curvilinear grid is proposed
- A WENO scheme is decomposed into consistent and dissipation parts
- Appropriate modification is adopted for both consistent and dissipation parts
- This technique works well for standard FDWENO and high resolution WENO-CU schemes
- Errors in inappropriate schemes are decomposed and discussed in detail